Python Activity-Based Learning
29. Simple Password Validation

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

Write a program to check a sample password. A valid password must have at least 8 characters and contain at least one digit.

2. Objectives

• Read a password string.

• Check its length.

• Check whether it contains a digit.

3. Assumptions

• The password is checked only for length and digits.

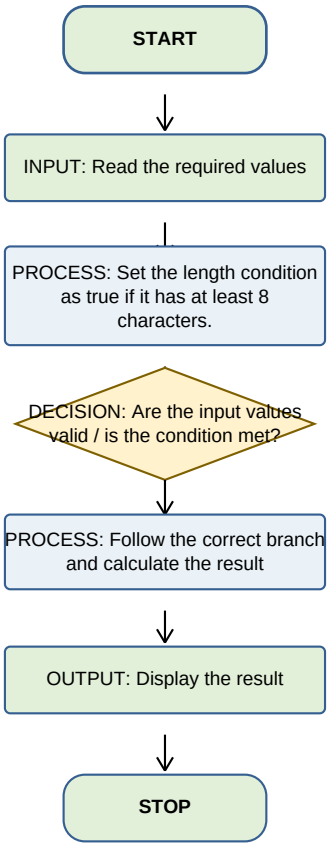
• The program does not save or display the password again.

4. Algorithm (Step by Step)

1. Start.
2. Read the password.
3. Set the length condition as true if it has at least 8 characters.
4. Check each character to see whether it is a digit.
5. If both conditions are true, display Valid password.
6. Otherwise, display Not valid and show the failed rule.
7. Stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT password
long_enough = length(password) >= 8
has_digit = FALSE
FOR each character in password
    IF character is a digit THEN has_digit = TRUE
IF long_enough AND has_digit THEN PRINT Valid
ELSE PRINT Not valid and missing rules
STOP
```

7. Python Source Code

```
password = input("Enter a sample password: ")

long_enough = len(password) >= 8
has_digit = False

for ch in password:
    if ch.isdigit():
        has_digit = True

if long_enough and has_digit:
    print("Valid password")
else:
    print("Invalid password")
    if not long_enough:
        print("Password must have at least 8 characters")
    if not has_digit:
        print("Password must contain at least one digit")
```

8. Test Cases

Input / Test	Expected result	What it checks
student123	Valid password	Length and digit condition pass
student	Invalid; needs 8 characters and a digit	Both conditions fail

Input / Test	Expected result	What it checks
hello123	Invalid; needs 8 characters	Only length condition fails

9. Sample Output

```
Enter a sample password: student123
Valid password
```

10. Conclusion

This program checks a password using string operations, a loop, and conditional statements.